

TempoRAG-KG: Temporal Knowledge Graph-Augmented Retrieval for Multi-Hop Question Answering

Progress Report

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Abstract

We report mid-project progress on **TempoRAG-KG**, a retrieval-augmented generation (RAG) system that couples a temporal knowledge graph with chunk-level retrieval to answer multi-hop, time-sensitive questions. During the first half of the project we piloted HotpotQA and MuSiQue as evaluation corpora, but found that their Wikipedia-anchored factoids do not reliably carry the kind of temporal structure our hypothesis targets. We therefore pivoted to a corpus of 25 SEC 10-K filings (5 technology mega-caps $\times$ 5 fiscal years, 2019–2024), producing 7,467 item-level chunks and a hand-vetted question bank of 129 items (50 home-grown + 79 multi-hop). On this corpus we have completed two of four retrieval conditions across 7 LLMs spanning 3 providers. Our first key finding is that a temporal year-mask applied on top of cosine retrieval (TimeFilter, L1) produces a *universal* positive lift over a vanilla baseline (L0), ranging from +5.6% to +24.7% in token-F1. However, the hop=3 slice barely moves (+0.007), which isolates the gap that our knowledge-graph conditions (L2, L3) are designed to close. KG extraction over the full corpus is in progress at the time of writing; L2 / L3 retrievers, ablations, and the final interaction analysis remain as the closing phase of the project.

1 Introduction

Retrieval-augmented generation (Lewis et al., 2020) has become a standard recipe for question-answering over document corpora, but off-the-shelf dense retrieval struggles when the *same entity* is described differently across time. A question such as “did Microsoft’s operating margin improve after the Activision acquisition?” requires the system to reason over facts that are true in one fiscal year and false in another — a property that flat chunk similarity is blind to.

A discovery-driven pivot. Our original proposal planned to evaluate on HotpotQA (Yang et al., 2018) and MuSiQue (Trivedi et al., 2022). During an EDA pass we inspected the temporal expressions present in those benchmarks and found two problems. First, the vast majority of multi-hop bridges are temporally static (“who was born in X?”), so even a perfect year filter cannot demonstrate lift. Second, Wikipedia text drifts: the “gold” supporting sentence may have been edited after the QA pair was written, which leaks noise into any retrieval-quality metric. This pushed us to look for a corpus where (a) each document is stamped with a machine-verifiable fiscal year, and (b) the same entity is described at multiple, non-overlapping points in time. SEC 10-K filings satisfy both criteria directly: each filing’s fiscal-year metadata is authoritative, and a 5-year window over 5 issuers gives 25 documents that repeatedly re-describe the same entities.

What this report covers. This progress report documents (i) the motivation for the pivot, (ii) the 10-K corpus and QA set we assembled, (iii) the pipeline we have built so far, and (iv) the first evaluation results from the L0 (vanilla cosine) and L1 (TimeFilter) conditions across seven LLMs. The two remaining conditions, L2 (KG²RAG-style graph walk) and L3 (TempoRAG-KG with temporal edges), depend on the running KG extraction and are discussed as in-flight work in Section 6.

Contributions to date.

1. A 7,467-chunk, item-level SEC 10-K corpus tailored to temporal multi-hop evaluation, with a 129-item hand-vetted QA set annotated for hop count and temporal scope.
2. A reproducible retrieval + evaluation harness (cached embeddings, cached generations, bootstrap CIs) supporting 7 LLMs $\times$ 3 providers on identical questions.

- Empirical evidence that a deterministic year-mask alone produces universal positive F1 lift, but leaves the hop=3 slice effectively flat — motivating the KG conditions that constitute the remainder of the project.

2 Related Work

Graph-based RAG. Using a knowledge graph to guide retrieval is an active area (Edge et al., 2024); our work builds directly on KG²RAG. Zhu et al. propose augmenting chunk retrieval with a graph walk over entity-linked seeds to widen the context available to the generator (Zhu et al., 2025). We adopt this as our L2 condition. Crucially, KG²RAG’s graph is untyped with respect to time: an edge either exists or does not. Our L3 condition extends the same retrieval signal with temporal validity intervals on each edge.

TempAgent and GoG. Several recent systems layer a planner over the graph to decide *which* edges to expand next (Xiong et al., 2025; He et al., 2024). Our design differs: we do not plan, we simply mask the candidate pool by fiscal year (L1) or by overlap of the question’s year scope with each edge’s validity window (L3). This keeps the experiment interpretable — any lift we observe is attributable to the temporal signal, not to emergent planner behaviour.

FinReflectKG-MultiHop. Kishore et al. recently released a multi-hop QA set over 10-K filings derived from a KG-constrained generator (Kishore and Collaborators, 2025). We use a subset of their questions (79 items) as an external anchor alongside our own 50 home-grown questions, so that our evaluation does not rely exclusively on questions we authored ourselves.

3 Research Questions

The experimental design is a 2×2 factorial over the two interventions the system combines: presence/absence of a *temporal filter* and presence/absence of a *knowledge-graph walk* (Table 1). The four cells correspond to the four pipeline conditions introduced in Section 4.3. All research questions below are stated against this matrix so that each one maps cleanly to a specific cell comparison.

RQ1 — Where does KG²RAG fail on temporal multi-hop QA? **IV:** graph walk on, temporal off (L2 cell, relative to L0). **DV:** per-slice token-F1

	No KG	+ KG
No Temporal	L0 Vanilla ✓	L2 KG ² RAG ✗
+ Temporal	L1 TimeFilter ✓	L3 TempoRAG-KG ✗

Table 1: 2×2 experimental design. ✓ = complete at the time of this report; ✗ = pending KG extraction.

(hop, scope). **H1:** KG²RAG improves over Vanilla on hop- ≥ 2 slices but leaves an identifiable *temporal* failure mode that a graph walk alone cannot close. *Pending KG extract.*

RQ2 — Does TempoRAG-KG lift token-F1 over KG²RAG? **IV:** temporal edges on / off with the KG held constant (L3 vs L2). **DV:** token-F1 on the 129-item QA set and on the hop- ≥ 2 slice. **H2:** L3 > L2 with the largest gain on slices where the question explicitly spans fiscal years (*inter_year*, *fiscal_vs_calendar*). *Pending.*

RQ3 — How accurate is the temporal KG extraction? **IV:** the extraction prompt (Appendix A) applied to each of the 7,467 chunks. **DV:** fraction of extracted triples whose `valid_from` / `valid_to` intervals agree with the source filing’s fiscal-year metadata. **H3:** temporal-interval accuracy is high enough that the L3 retriever’s ceiling is bounded by graph coverage (a design question), not by spurious intervals (a data question). *Measured once the full extract completes.*

RQ4 (★) — Can a small LM with temporal grounding approach a large LM without it? **IV:** model scale crossed with retrieval condition (the seven models in Table 3 $\times$ L0–L3). **DV:** the gap between the best small-open-model (llama-3.1-8B or gpt-oss-20B) at L3 and the best large-closed-model (gpt-4o) at L0. **H4:** temporal grounding substantively narrows this gap — i.e. capability and retrieval are partially substitutable for temporal QA. This is the headline question for the project and the one our design is optimised around.

Current state. Section 5 reports the L0 vs L1 comparison, which is not by itself sufficient to answer any of RQ1–RQ4 but does supply one necessary input: the size and uniformity of the lift that a purely temporal signal, with no KG, can produce on this corpus. The results are consistent with RQ4’s hypothesis in the directional sense (small models benefit at least as much as large models from the temporal filter), but a definitive answer requires the L2 and L3 cells.

4 Methodology

4.1 Corpus

We downloaded 10-K filings directly from the SEC EDGAR API (data.sec.gov/submissions) using a custom rate-limited (5 req/s) downloader (`scripts/download_10k.py`) and verified each filing with its SHA-256 digest. The core evaluation corpus uses the five technology mega-caps — AAPL, MSFT, GOOGL, AMZN, META — across fiscal years 2019–2024 (25 filings). We segment each filing into item-level chunks (Items 1, 1A, 7, 7A, 8) of approximately 1,000 tokens, producing 7,467 chunks. All chunks are embedded once with `text-embedding-3-small` and cached (total embedding spend: approximately \$0.07).

4.2 QA Set

Our 129-item QA set is assembled from two sources:

- 50 *home-grown* questions authored by the team and cross-reviewed for factual correctness against the underlying filings.
- 79 questions drawn from FINREFLECTKG-MULTIHOP, filtered to those whose gold answer span is present in our 25-filing corpus.

Every question is annotated with a hop count and a scope label (*intra*, *inter_year*, *cross_company*, *fiscal_vs_calendar*, *forward_looking*). Table 2 summarises the scope distribution for the home-grown half; the multi-hop half is scope-mixed and dominated by *hop* = 2 items ($n = 104$).

Scope	Count
<i>cross_company</i>	18
<i>inter_year</i>	15
<i>intra</i>	10
<i>fiscal_vs_calendar</i>	5
<i>forward_looking</i>	2
Total (home-grown)	50

Table 2: Scope distribution of the 50 home-grown questions. Scope is an orthogonal axis to hop count: a *cross_company* question may be hop 2 or hop 3 depending on how many filings are required to answer it.

4.3 Pipeline

Figure 1 shows the full pipeline: 25 filings are chunked, embedded once, and fed to four retrieval branches that share a single answer-generation stage. The two completed branches are:

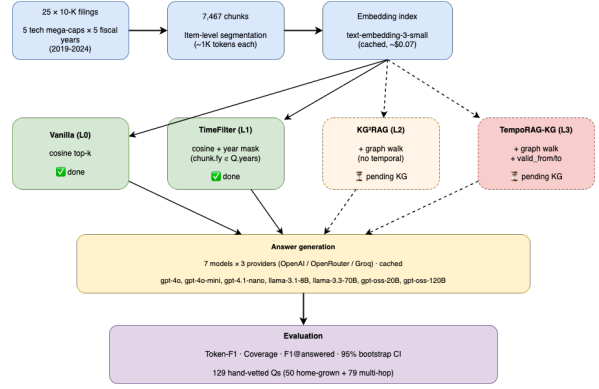


Figure 1: End-to-end pipeline. Blue: corpus preparation (done). Green: completed retrieval branches (L0, L1). Orange/red dashed: pending retrieval branches (L2, L3). Yellow: shared answer-generation stage. Purple: evaluation.

- **L0 Vanilla** — top- k cosine retrieval over all 7,467 chunks (retrieve in `src/retrieval.py`).
- **L1 TimeFilter** — same retrieval, but scores for chunks whose fiscal year is not in the question’s year set are masked to $-\infty$ (`retrieve_with_year_filter` in the same module).

If the masked pool contains fewer than k chunks, we fall back to the unfiltered pool to avoid empty contexts. The two pending branches (L2, L3) share the same generator and prompt and differ only in how the candidate pool is assembled.

4.4 Metrics

We report **Token-F1** with SQuAD-style normalisation, **Coverage** (the fraction of rows where the model attempted an answer rather than refusing), and **F1@answered** (token-F1 restricted to the answered subset). All intervals are 95% non-parametric bootstrap CIs with $n = 1000$ and a fixed seed (the implementation is `src/eval.py:aggregate` and `src/eval.py:bootstrap_ci`). A refusal or empty string is detected by the `is_idk` regex in the same module and counted as not-answered rather than as a wrong answer.

4.5 Models

We run the same 129 questions through seven LLMs spanning three providers so that any finding generalises across both scale and vendor (Table 3).

Cost and caching. Each question $\times$ model $\times$ condition produces exactly one generation, which

Model	Provider	Role
gpt-4o	OpenAI	large, closed
gpt-4o-mini	OpenAI	mid, closed
gpt-4.1-nano	OpenAI	small, closed
llama-3.1-8B	Groq	small, open
llama-3.3-70B	Groq	large, open
gpt-oss-20B	OpenRouter	small, open
gpt-oss-120B	OpenRouter	large, open

Table 3: The seven LLMs used as answer generators. All receive the same retrieved context and the same prompt; only the model identity varies.

is written to an on-disk cache keyed by prompt and generation parameters; subsequent runs are free. The open-model rows (llama-3.1-8B, llama-3.3-70B on Groq; gpt-oss-20B/120B on OpenRouter) use free-tier endpoints; the three OpenAI models together produced $\approx 1,800$ cached generations for the L0 and L1 sweeps at a total spend well under the project’s \$20 budget. Cumulative generation wall-clock across all 7 models is ≈ 2.2 hours.

5 Results

This section reports the state of the evaluation at the time of writing. Only the L0 and L1 conditions are complete; L2 and L3 rows will be added once the KG extraction finishes and the L2/L3 retrievers are implemented.

5.1 Overall

Across all seven LLMs, the L1 year-mask strictly dominates the L0 vanilla baseline in token-F1 (Table 4). The lift ranges from +5.6 % (gpt-oss-120b) to +24.7 % (llama-3.1-8B), and coverage also rises under L1, confirming that the mask is not simply trading off retrieval precision for refusal rate.

Retrieval, not answer quality, is the bottleneck. Comparing L0 and L1 on the two sub-metrics makes this more precise. L1 coverage rises by +3.1 pp (gpt-oss-120b) to +10.1 pp (llama-3.3-70B) — i.e. the mask causes the retrieved context to contain an answerable span more often. But F1@answered — the token-F1 *conditional* on the model having produced a non-refusal — barely moves between L0 and L1 (for example, gpt-4o goes from 0.385 to 0.391, gpt-4o-mini from 0.396 to 0.395). This pattern tells us that, at current retrieval quality, each model’s answer-writing ability is already close to ceiling on whatever evidence it is given; the further F1 lifts we are chasing must come from better retrieval (KG walk, temporal edges),

not from a larger generator.

5.2 By hop count

To see whether the year-mask alone is enough to handle multi-hop questions, we slice by hop count for a representative small model (gpt-4.1-nano, Table 5). Two things stand out. First, the hop=2 slice (the bulk of the QA set) improves by +0.048 F1. Second — and this is the main motivating observation for the KG conditions — the hop=3 slice moves by only +0.007, i.e. the temporal signal on its own is effectively blind to the genuinely multi-hop cases. This is the gap the L2 / L3 graph walk is designed to close.

5.3 By temporal scope

Table 6 stratifies the same gpt-4.1-nano run by the scope label. The mask delivers its largest lift exactly where a year mask *should* help: on *inter_year* questions that span multiple fiscal years (+0.055, $n = 53$) and on *cross_company* questions that still pin a year (+0.051, $n = 24$). Two slices where the mask cannot help are also visible. On *intra* (same-year) questions the mask is essentially a no-op ($\Delta = -0.001$, $n = 45$): the question and all candidate chunks already share one year, so removing other years changes nothing. On *fiscal_vs_calendar* ($n = 5$) the mask actually regresses by -0.079 , an expected failure mode: a question of the form “in Q3 2022...” has a calendar-year stamp, but the answer lives in a filing stamped with a fiscal year that may differ, and the hard mask removes the correct filing. The sample is small ($n = 5$) but the direction is the one we predicted in Section 3.

6 Progress Summary

Completed. Corpus preparation (25 filings, 7,467 chunks, cached embeddings), QA-set construction (129 items, hop/scope annotated), retrieval harness (L0, L1), evaluation harness (Token-F1, Coverage, F1@answered, bootstrap CI), answer-generation cache across 7 models $\times$ 3 providers, and the progress-report slide deck.

In flight. KG extraction over the full corpus is currently running. Each chunk is passed through a structured-extraction prompt that yields typed triples with explicit `valid.from / valid.to` intervals; this is the input substrate for both the L2 and L3 conditions.

Remaining work. A light filter pass on the extracted triples (drop trivially-true relations, fix the

Model	L0 token-F1		L1 token-F1		Δ %	L1 Coverage	L1 F1@ans
	mean	95% CI	mean	95% CI			
gpt-4.1-nano	0.229	[0.194, 0.270]	0.258	[0.221, 0.296]	+12.7	70.5%	0.366 [0.333, 0.398]
gpt-4o	0.183	[0.141, 0.223]	0.221	[0.184, 0.264]	+20.9	56.6%	0.391 [0.348, 0.432]
gpt-4o-mini	0.230	[0.193, 0.274]	0.248	[0.206, 0.290]	+7.8	62.8%	0.395 [0.357, 0.435]
gpt-oss-20B	0.120	[0.088, 0.159]	0.145	[0.110, 0.184]	+21.0	42.6%	0.336 [0.279, 0.391]
gpt-oss-120B	0.131	[0.094, 0.172]	0.138	[0.103, 0.181]	+5.6	38.8%	0.356 [0.296, 0.418]
llama-3.1-8B	0.145	[0.114, 0.180]	0.180	[0.147, 0.216]	+24.7	57.4%	0.314 [0.276, 0.355]
llama-3.3-70B	0.145	[0.112, 0.181]	0.179	[0.144, 0.215]	+23.7	53.5%	0.335 [0.294, 0.373]

Table 4: Per-model token-F1 under L0 (vanilla cosine) and L1 (TimeFilter), with 95 % non-parametric bootstrap CIs ($n = 1000$, fixed seed). Relative lift, L1 coverage, and L1 F1@answered (with CI) are shown on the right. The L1 mean is above the L0 mean for every model; no L1–L0 delta flips sign inside its CI. L2 and L3 rows will be added when those conditions complete.

Hop	n	L0	L1	Δ
1	10	0.337	0.206	−0.131
2	104	0.234	0.281	+0.048
3	15	0.123	0.131	+0.007

Table 5: gpt-4.1-nano token-F1 by hop count. The hop=1 regression is a known artefact: some hop=1 questions carry a year label that is tighter than the span the filing actually reports the answer in, so the hard mask removes the relevant chunk. A softer temporal weighting is a natural mitigation but is outside the scope of this report.

Scope	n	L0	L1	Δ
inter_year	53	0.220	0.276	+0.055
cross_company	24	0.264	0.315	+0.051
forward_looking	2	0.138	0.157	+0.019
intra	45	0.216	0.216	−0.001
fiscal_vs_calendar	5	0.301	0.222	−0.079

Table 6: gpt-4.1-nano token-F1 by scope. *forward_looking* ($n = 2$) is not statistically meaningful and is included only for completeness; the *fiscal_vs_calendar* regression is discussed in the text.

comparative-value extraction bug); implementation of the L2 (KG²RAG-style) and L3 (TempoRAG-KG) retrievers; identical 7-model sweeps for L2 and L3; the interaction analysis required by RQ3; and the write-up.

6.1 Limitations at this stage

Four threats to validity deserve to be named explicitly before the final report. **(i)** The 129-item QA set is hand-vetted and internally consistent but small; individual scope cells (especially *forward_looking*, $n = 2$, and *fiscal_vs_calendar*, $n = 5$) have too few items to support confident sub-claims on their own. **(ii)** The current results cover only the L0/L1 axis of the 2×2 design; any claim about the KG

contribution or about the interaction between KG and temporal signal is *not* supported by this report. **(iii)** L1 uses a *hard* year mask: the regression on *fiscal_vs_calendar* (Table 6) is a direct consequence of that design choice, and a softer temporal weighting may recover that slice. **(iv)** The corpus is 25 filings from five issuers in one sector; findings should not be extrapolated to financial QA in general without further evaluation.

7 Conclusion

The evidence we have at the time of this report is narrow but clear: on a 10-K corpus where every document carries an authoritative fiscal-year label, a deterministic year-mask applied on top of a cosine retriever raises token-F1 for every one of the seven LLMs we tested, with no observed reversals. The same experiment also shows that this temporal signal, on its own, essentially does not help on the hop=3 slice. We read this as a clean separation of the two effects our system is designed to combine: the L1 result establishes that temporal grounding carries measurable signal on this corpus, and it leaves the multi-hop gap open for the KG conditions to address. Importantly, the L1 lift is as large for the two small open models (llama-3.1-8B, gpt-oss-20B) as it is for the large closed models, which is a directional signal in favour of RQ4 (Section 3). The headline test — whether a small LM at L3 approaches a large LM at L0 — remains the next and decisive milestone for the project.

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A Extraction Prompt (v1)

See `prompts/extract_v1.txt` in the repository. A verbatim copy will be included in the final report.

B Answer Prompt

See `prompts/answer_v1.txt`. The prompt is deliberately minimal: it supplies the retrieved chunks, the question, and an instruction to refuse

if the answer is not present in the context. Refusal phrasing is what the `is_idk` regex in `src/eval.py` is tuned to recognise.

C Scope Taxonomy

intra Answer is contained within a single (filing, item) pair.

inter_year Answer requires comparing two or more fiscal years of the same issuer.

cross_company Answer requires comparing two or more issuers, possibly in a single year.

fiscal_vs_calendar Question references a calendar-year event (“Q3 2022”) that must be mapped onto the issuer’s fiscal year.

forward_looking Answer is a projection or guidance statement rather than a realised value.